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Differentiation of human intestinal organoids with endogenous vascular endothelial cells

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SUMMARY

Human pluripotent stem cell (hPSC)-derived intestinal organoids (HIOs) lack some cellular populations found in the native organ, including vasculature. Using single cell RNA sequencing (scRNA-seq), we have identified a population of endothelial cells (ECs) present early in HIO differentiation that declines over time in culture. Here, we developed a method to expand and maintain this endogenous population of ECs within HIOs (vHIOs). Given that ECs possess organ-specific gene expression, morphology and function, we used bulk RNA-seq and scRNA-seq to interrogate the developing human intestine, lung, and kidney in order to identify organ-enriched EC-gene signatures. By comparing these gene signatures and validated markers to HIO ECs, we find HIO ECs grown *in vitro* share the highest similarity with native intestinal ECs relative to kidney and lung. Together, these data demonstrate HIOs can co-differentiate a native EC population that are properly patterned with an intestine-specific EC transcriptional signature *in vitro*.

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Author contributions

EMH and JRS conceived the study. JRS supervised the research. EMH developed tissue dissociation methods and isolated primary ECs for bulk RNA sequencing. AW, YHT, EMH, MC developed tissue dissociation methods and generated single cell RNA sequencing data. JW and MC performed computational analysis. MC, JW, EMH and JRS interpreted computational results. EMH, CS, and AES performed IF and FISH experiments and imaging. EMH, CWS, AES, and AW, and YHT processed primary tissue for IF/FISH. EMH, SH, and MMC generated human intestinal organoids. IG provided critical material resources for this work. EMH assembled figures. EMH and JRS wrote the manuscript. EMH, JW, AW, and MC contributed methods. All authors edited, read and approved the manuscript.

Competing interests

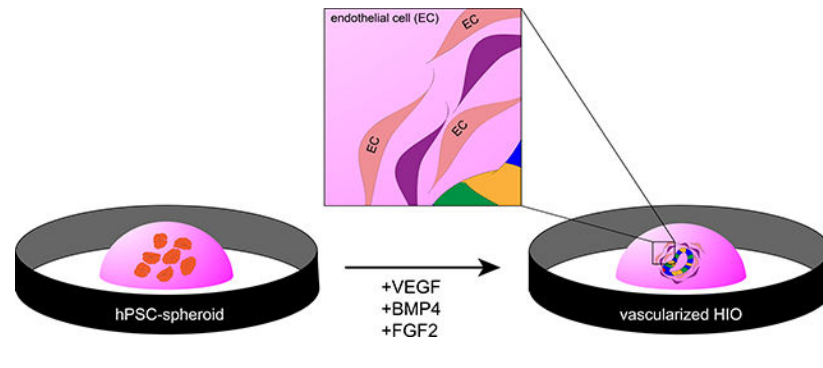
EMH and JRS hold a provisional patent application pertaining to the vHIOs described herein.

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eTOC blurb

Holloway et al. investigate cellular heterogeneity during hPSC-derived intestinal organoid development and identify resident endothelial cells (ECs). ECs are usually lost over time but can be expanded using modified media conditions. *In vivo* human organ-specific EC signatures reveal that organoid ECs have the highest transcriptional similarity to native intestinal ECs.

Graphical Abstract



INTRODUCTION

The development of human pluripotent stem cell (hPSC) derived small intestinal organoids (HIOs) (Spence et al., 2010) or primary organoids derived from human donor tissues (Sato et al., 2011), has led to an improved understanding of human intestinal development and physiology. In order to generate hPSC-derived HIOs, recombinant proteins and/or small molecules are exogenously supplied to cultures in order to mimic the *in vivo* signaling environment present during development, through a process called directed differentiation (McCracken et al., 2011). Current methods to differentiate HIOs generate both endodermal cells that will give rise to the HIO epithelium as well as a mesoderm population that can give rise to cells of the lamina propria, such as fibroblasts and smooth muscle cells (Finkbeiner et al., 2015a, 2015b; McCracken et al., 2011; Spence et al., 2010; Watson et al., 2014; Wells and Spence, 2014). However, HIOs do not fully recapitulate the complexity of the native human intestine, lacking cellular components including immune cells, enteric neurons (Workman et al., 2016), vasculature and the microbiome (Hill et al., 2017a; Leslie et al., 2014).

Co-culture and transplantation approaches have been developed to increase the complexity of HIOs (Holloway et al., 2019). For example, hPSC-derived enteric neural lineages have been added *in vitro* and further matured *in vivo* to establish a functional enteric nervous-like system within HIOs (Schlieve et al., 2017; Workman et al., 2016). Similarly, microinjection of *E. coli* into the HIO lumen has permitted the study of epithelial response to early gut colonization (Hill et al., 2017a, 2017b). However, vascularization of HIOs has been restricted to *in vivo* models, whereby HIOs are transplanted into highly vascularized regions of immunocompromised mice (Cortez et al., 2018; Watson et al., 2014). In these environments, HIOs undergo extensive vascularization by the murine host tissue, and increase in complexity to resemble mature intestinal tissue (Finkbeiner et al., 2015b; Watson

et al., 2014). However, co-culture approaches or co-differentiating HIOs with a native vasculature prior to *in vivo* engraftment has not yet been achieved.

Here, we performed single cell RNA sequencing (scRNA-seq) at various timepoints across HIO differentiation *in vitro* and observed a population of endothelial-like cells (ECs) present within HIOs early during differentiation; however, these cells are not well maintained during prolonged culture under standard growth conditions. This suggested that early during HIO differentiation, cells within the culture are capable of giving rise to EC-like cells. Based on these observations, we hypothesized that a modified directed differentiation approach would allow the expansion and maintenance of a more robust EC population within HIOs (termed vHIO). Our findings demonstrate that modified culture conditions result in an approximate 9-fold increase in EC-like cells within HIOs without dramatically impacting the other HIO cell populations present (i.e. epithelium, mesenchyme), and support the survival of this population of ECs within HIOs in culture for months.

Since organ-specific morphology in vascular beds has long been appreciated (Aird, 2007), and organ-specific transcriptional signatures and functions have been described in mouse (Ding et al., 2011; Kalucka et al., 2020; Lee et al., 2014; Nolan et al., 2013) and human tissues (Marcu et al., 2018), we further sought to determine if HIO ECs were properly patterned by interrogating human fetal intestine, lung, and kidney tissue to identify and validate organ enriched EC gene signatures and individual genes. These data were then used to assess the extent to which HIO ECs resembled human intestinal ECs. After two months of culture, we observed that relative to human fetal intestine, lung and kidney ECs, HIO ECs were the most transcriptionally similar to the fetal intestine ECs, suggesting that HIOs possess intrinsic properties sufficient to induce ECs to undergo proper organ-specific patterning.

Taken together, this work shows that a native EC population is present within HIO cultures, and that these EC-like cells can be maintained under defined culture conditions (vHIOs) for prolonged culture periods. We further present human intestinal, lung and kidney EC data sets that can be used as a reference for comparison against *in vitro* derived organoids and specific cell types found within *in vitro* organoids. Using transcriptional profiles and a panel of several validated markers, we conclude that HIO ECs undergo organ-specific patterning *in vitro*, and most closely resemble an intestinal EC population.

RESULTS

hPSC-derived human intestinal organoids (HIOs) develop a population of endothelial cells (ECs)

Human intestinal organoids (HIOs) have been almost exclusively characterized following growth for several weeks in culture (Capeling et al., 2019; Finkbeiner et al., 2015b; Spence et al., 2010; Tsai et al., 2017). Through these analyses, various intestinal epithelial and mesenchymal populations have been identified. Initial observations suggested that HIOs lacked an EC population (Spence et al., 2010); however, a thorough investigation of cellular heterogeneity within HIOs across time has not been carried out.

In order to gain insights into cellular heterogeneity during HIO growth and differentiation, we performed a single cell RNA sequencing (scRNA-seq) time course analysis. Analysis was carried out on HIO samples following hindgut patterning and specification into a CDX2+ intestinal lineage, the time when monolayer cultures give rise to 3-dimensional (3D) spheroids (herein referred to as 'day 0'), and several time points after embedding spheroids in Matrigel for growth and expansion into HIOs (days 3, 7, and 14) (Figure 1A). A total of 13,289 cells from these four time-points (days 0, 3, 7, 14) were included in the computational analysis and the data were visualized using UMAP dimensional reduction (Becht et al., 2019; Wolf et al., 2018) (Figure 1B). Cell classifications were carried out using canonical markers for lineages (i.e. epithelial, mesenchymal, endothelial) (Figure 1B, S1). We identified expected epithelial and mesenchymal cell lineages within HIOs; however, we also identified cell clusters possessing endothelial (cluster 14) and neuronal (clusters 11,15) gene signatures (Figure S1). The endothelial cluster was defined by a gene set expected to be present in ECs, including *CDH5*, *KDR*, *FLT1*, and *ESAM* (Figure 1C, S1). The time course revealed that the EC-like population in cluster 14 was most prominent early in HIO development, with the highest proportion of EC-like cells on day 3 of culture (3.30% of all day 3 cells), after which EC prevalence progressively decline to 0.73% on day 7 and 0.44% on day 14 (Figure 1D). Wholemout staining of day 4 HIOs with the EC lineage marker CD144 confirmed the existence of an EC-like population early during HIO growth (Figure 1E). Together, these data suggest that HIO differentiation cultures are capable of spontaneously differentiating EC-like cells that diminish over time.

Developing a method to expand and maintain ECs within HIOs

The progressive reduction of EC-like cells over time suggested that standard HIO culture conditions do not support robust long-term EC survival. Therefore, we sought to define culture conditions that would expand and maintain the native EC population that would also retain anterior-posterior patterning of the HIO epithelium, which is most similar to duodenum when grown under the standard conditions used here (Figure 1). We hypothesized that addition of growth factors previously shown to be important for vascular induction and maintenance in hPSC differentiations would improve EC differentiation and maintenance within HIOs, including VEGF, FGF2 (bFGF), and BMP4 (Orlova et al., 2014; Patsch et al., 2015; Sriram et al., 2015; Wimmer et al., 2019). However, while BMP4 and FGF2 are important for EC induction, they have been shown to influence anterior-posterior patterning of early endoderm lineages (Ameri et al., 2009; Munera et al., 2017; Serls et al., 2005). Therefore, attempts to influence ECs within the HIO were implemented following 3 days of spheroid growth in standard HIO media (EGF/NOG/RSPO – 'ENR'), since NOG is critical for anteriorization of the HIOs (Munera et al., 2017). The exception to this was the addition of VEGF on day 2 of ENR culture in an attempt to promote survival of endogenous ECs (Figure 2A). After 3 days of HIO patterning into an proximal/duodenal identity, HIOs were maintained in 'EGF-basal' growth media (without RSPO or NOG), as EGF is sufficient to support HIO growth following proximal-distal patterning (Munera et al., 2017). In order to induce and/or support ECs after duodenal patterning in ENR, we added combinations of VEGF, FGF2 (bFGF), and BMP4 to EGF-basal media and used qRT-PCR to screen for conditions that enhanced *CD31* and *CD144* mRNA expression (Figure S2A–D). The combination of VEGF, BMP4 and FGF2 from day 3 to day 6 resulted in the highest

expression of the pan-EC markers *CD31* and *CD144* when added to EGF-basal media for 3 days (Figure S2A–B). Longer term culture supplementation of EGF-basal media with VEGF supported the sustained increased expression of *CD31* and *CD144* relative to EGF-basal only controls and relative to all other growth factor combinations tested, while culture conditions containing FGF2 resulted in induction of anterior foregut marker *SOX2* (Figure S2C–D). Insights from the growth factors screens shaped the HIO with EC differentiation paradigm, herein referred to the vascularized HIO (vHIO) protocol.

The vHIO protocol involves supplementing the final day of ENR (day 2 of 3D spheroid culture) with VEGF (50 ng/ml) followed by treating HIOs for three days with EGF (100 ng/ml), VEGF (50 ng/ml), FGF2 (bFGF) (25 ng/ml), and BMP4 (25 ng/ml) (Figure 2A). vHIOs were then maintained in media supplemented with EGF (100 ng/ml) and VEGF (25 ng/ml) for up to two months prior to analysis, while controls were maintained in media containing EGF only (Figure 2A). To assess the proportion of EC-like cells in HIOs following the vHIO protocol, flow cytometric analysis of CD31+/CD144+ cells was used to compare control HIOs and vHIOs. After several weeks in culture, ECs constituted an average of $0.31\% \pm 0.26\%$ in control HIOs, whereas vHIOs possessed $3.17\% \pm 1.14\%$ CD31+/CD144+ (Figure 2B, Figure S2E). Identification of CD31+/CD144+ cells within vHIOs using immunofluorescence confirmed EC enrichment as double-positive cells were abundant and were observed within HIO mesenchyme (Figure 2C, S2G). EC enrichment in vHIOs persisted for two months in culture, the longest time point examined (Figure 2D,E, S2F).

To further support our flow cytometric and immunohistochemical data showing EC-enrichment, and to assess the effect of the vHIO protocol on overall HIO development and cellular makeup, scRNA-seq was performed on control and vHIO tissue after two months in culture. A total of 16,253 cells (8,973 control HIO and 7,280 vHIO) were analyzed and the data were visualized using UMAP dimensional reduction (Figure 2D–E). Epithelial lineages (clusters 2 and 3), mesenchymal lineages (clusters 0,1, and 4), and endothelial lineages (cluster 5) were classified based on expression of canonical makers (Figures 2D and S3A–B). Quantification of the contribution of control and vHIO cells to each cluster was performed (Figure S3D–F). ECs were enriched in the vHIO condition compared to control, comprising 2.03% of all cells sequenced in the vHIO condition, a 3.5-fold increase over control HIOs (Figure 2E). More generally, this scRNA-seq analysis demonstrated that all of the cell populations visualized by UMAP have contributions from both HIOs and vHIOs, showing that the vHIO differentiation paradigm does not result in gain or loss of epithelial or mesenchymal cell populations (Figure S3). To investigate subtle changes in cell composition that may result from the vHIO protocol, each cluster was computationally extracted and re-clustered (Figure S3G). Between 3–6 subclusters were predicted for each parent cluster, and relative contributions of control and vHIO to each subcluster were determined for each parent cluster (Figure S3G–J); however, we note that the boundaries between subclusters were not sharp, suggesting that gene expression variability in subclusters was not sufficient to resolve distinct sub-populations. These analyses revealed only subtle changes in cell composition between the two groups, demonstrating that the distribution of cells from HIOs or vHIO was largely within a range 2-fold in either direction (increased or decreased in vHIO relative to HIO controls). The lone exception was the extracted parental mesenchymal cluster 0, which had 5 subclusters (Figure S3G). Parental cluster 0, subcluster 1 showed a

~2-fold over-representation of cells derived from vHIOs relative to HIOs (Figure S3I–J). These results reinforce the conclusion that vHIOs have an enrichment of EC-like cells without dramatically influencing epithelial or mesenchymal diversity in the organoids.

The importance of EC-organ signaling during the development of several endoderm-derived organs including liver (Lammert et al., 2003; Matsumoto et al., 2001), pancreas (Kao et al., 2015; Lammert, 2001), and lung (Daniel et al., 2020; Lazarus et al., 2011; Vila Ellis et al., 2020) has been demonstrated. As such, we were interested in characterizing cellular crosstalk occurring between ECs and mesenchymal or epithelial lineages in HIOs. To do this, we performed *in silico* ligand-receptor analysis in the combined control HIO and vHIO 59d scRNA-seq dataset using CellphoneDB (Efremova et al., 2020). We found that ECs (Cluster 5) have the highest number of predicted interactions with HIO mesenchyme (Clusters 0,1,4) (Figure 3A). When exploring significant interactions involving HIO ECs, we found that ECs express high levels of ligands *PGFRB*, *HBEGF*, and *DLL4* with corresponding receptor expression in HIO mesenchyme (*PDGFRB*, *EGFR*), epithelium (*EGFR*), and ECs (*NOTCH4*), respectively (Figure 3B–D). Additionally, HIO mesenchyme expresses angiogenic cytokine CXCL12, with ECs robustly expressing its receptor CXCR4 (Figure 3B–D). These results suggest that reciprocal signaling is occurring among ECs and other HIO lineages and allow us to begin developing hypotheses about cell-cell crosstalk that may be important for survival and patterning. Validating these findings *in situ* will be an important next step.

Defining a human intestinal EC transcriptional signature

Organ-specific properties including morphology, transcriptional signatures, and function have been described in mouse and human organs (Daniel et al., 2018; Ding et al., 2010, 2011; Lee et al., 2014; Marcu et al., 2018; Nolan et al., 2013); however, we sought to better understand intestine-specific EC properties in order to compare *in vivo* ECs with ECs found within HIOs. To identify intestine-specific EC transcriptional signatures, we performed bulk RNA sequencing (RNAseq) on FACS-isolated EC (CD31+/CD144+) and non-EC (CD31-/CD144-) populations from human fetal intestine, lung, and kidney spanning 11–20 weeks of development (Figure 4A). Bulk RNAseq data demonstrated that canonical EC markers (*CD31*, *CD144*, *KDR*) were enriched in the CD31+/CD144+ isolated populations (Figure 3B). Additionally, the EC samples had low-to-no expression of non-endothelial mesenchyme genes (i.e. *PDGFRa*), hematopoietic genes (i.e. *CD45*), and epithelial genes (i.e. *EPCAM*), which were expressed in the double negative (CD31-/CD144-) population as expected (Figure 4B). Unsupervised hierarchical clustering of all samples revealed that primary ECs from different human organs (intestine, lung, kidney) formed their own clade, and the CD31-/CD144- ‘non-ECs’ formed another clade. Notably, biological replicates of the same organ were closest in similarity compared to EC samples isolated from the other organs, to non-ECs, and to human umbilical vein endothelial cells (HUVECs), suggesting that there are organ-specific transcriptional profiles among human fetal ECs (Figure 4C). Principal component analysis of the primary ECs produced organ-specific clustering, further supporting the existence of organ-specific transcriptional differences across human fetal intestine, lung, and kidney ECs (Figure 4D). K-means (k=20) gene clustering was used to identify genes enriched within the ECs of a single organ. At this resolution, pan-EC enriched

gene clusters (0, 1, 2, 3, 4) as well as organ-specific EC-enriched gene clusters (6, 8, 17) were identified (Figure 4E). Over 100 organ-specific EC-enriched genes were computationally identified for all 3 organs. These gene lists are herein referred to as the lung EC signature (IECs), intestine EC signature (iECs), and kidney EC signature (kECs) (Supplemental Table 1).

Validation of the organ-specific EC genes and gene signatures identified by bulk RNAseq (Figure 4) was carried out using both scRNA-seq and fluorescent *in situ* hybridization (FISH) (Figure 5). We used scRNA-seq to profile human fetal lung (Miller et al., 2020), intestine (Czerwinski et al., 2020), and kidney across 7 specimens spanning 13.5–19 weeks of gestation (Figure S4A). In total, our analyses included 62,046 cells across these three organs (Figure 5A and S4A). An EC cluster was identified for each organ using cell type scoring (see Methods), which leverages gene cohorts canonically associated with different cell classes (i.e. epithelium, mesenchyme, endothelium, immune, neuronal) (Miller et al., 2020) (Figure 4A, S5A). Based on this analysis, EC clusters were computationally extracted and re-clustered; the EC clusters collectively contained 3,082 cells and are comprised of 1,361 lung ECs, 877 intestine ECs, and 844 kidney ECs (Figure 5B, Figure S4A). This analysis also revealed EC heterogeneity in the form of sub-clusters, including a lymphatic EC cluster apparent in the lung (cluster 5) and intestine (cluster 2) as defined by *PROX1* expression (De Val and Black, 2009; Wigle and Oliver, 1999; Wigle et al., 2002) (Figure S4B). Organ-specific EC genes identified in bulk RNAseq data were validated in scRNA-seq data (Figure 5C, E, and G, Figure S4C–E). The single cell data showed that organ-specific EC genes are expressed by vascular ECs in a manner expected based on the bulk sequencing data. Notably, despite being enriched in ECs in an organ-specific manner, some markers were also expressed by non-EC lineages among the organs profiled (Figure S5). Collectively, these data showed that a given gene is enriched in ECs of a single organ relative to the other organs, but that expression may not be exclusive to ECs (Figure S5). We also used multiplexed FISH to validate a cohort of organ-specific EC genes. Sample matched human fetal lung, intestine, and kidney were stained for the panel of organ-specific EC markers alongside the pan-EC marker *CDH5* (Figure 5D, F, H and S4). Three EC-specific organ-enriched genes were selected from lung (*CA4*, *ADRB1*, *VIRP1*), intestine (*MEOX1*, *NKX2.3*, *FABP4*) and kidney (*CRHBP*, *IRX3*, *IRX5*) for validation by multiplexed FISH or immunofluorescence (Figure 4E, Figure 5, Figure S4C–E). This approach confirmed data obtained in bulk and single cell RNA sequencing. Through this validation effort, a panel of 9 organ-specific EC enriched markers across human fetal lung, intestine, and kidney can be used in combination with computational analysis to assess the patterning of ECs co-differentiated within HIOs.

HIO ECs share the highest transcriptional similarity with fetal intestinal ECs

Given the unique organ-specific gene expression by ECs *in vivo* during human development, we wanted to assess the extent of organ-specific patterning of the ECs within HIOs. The EC cluster from scRNA-seq (cluster 5, Figure S3A–B) of 59d control and HIO ECs was computationally extracted, reclustered and visualized using UMAP (Figure 6A). To determine how similar gene expression of HIO ECs was to native organ gene signatures, organ-specific EC-enriched gene signatures (IECs, iECs, and kECs) identified in bulk

RNAseq (Figure 4) served as the foundation for organ-specific cell type scoring analysis. The HIO ECs were analyzed for expressed of the most highly enriched genes found in IECs, iECs, and kECs (see Methods). As such, the expression of 137 IECs, 89 iECs, and 50 kECs genes was determined in HIO ECs, and the average expression for each organ-signature was calculated for HIO ECs. Based on this analysis ECs co-differentiated within HIOs had the highest average expression of the intestinal EC signature relative to lung and kidney ECs (Figure 6B,C). FISH analysis of 29-day vHIOs confirmed expression of the intestine-specific marker, *NKX2.3* by HIO ECs (Figure 6D); and a lack of detectable expression of markers validated for the lung or kidney (Figure S5). Notably, the intestinal EC marker *MEOX1* was not detectable by FISH in vHIOs, suggesting that intestinal patterning may be incomplete in HIO derived ECs at this early time point (Figure S6).

DISCUSSION

While organoids are genetically tractable, complex *in vitro* human systems, they do not entirely recapitulate the full complement of cell types or complex physiology found in native tissues (Holloway et al., 2019). Several groups have been working to improve organoids, and approaches include implementing strategies to increase organoid complexity and maturation to more accurately mimic the native tissue (Fujii et al., 2018; Low et al., 2019; Mansour et al., 2018; Ouchi et al., 2019). In the case of cell types lacking in HIOs, such as ectoderm-derived enteric neurons, co-culture techniques using enteric neuron precursors (vagal neural crest cells) have been developed, facilitating the integration of a functional enteric nervous system that can stimulate motility following transplantation (Schlieve et al., 2017; Workman et al., 2016). While HIO vascularization has been achieved following *in vivo* transplantation by the host vasculature, developing a native human vasculature within HIOs has been elusive. Co-culturing endothelial cells (ECs) has proven successful in other *in vitro* systems, including hPSC-derived hepatic endoderm cultured with primary and hPSC-derived ECs leading to complex liver bud organoids with improved hepatocyte function (Camp et al., 2017; Takebe et al., 2013, 2017). A unique aspect of hPSC-derived HIOs is the co-differentiation of both intestinal epithelium and mesenchyme, which gives rise to diverse mesenchymal lineages including smooth muscle, myofibroblasts, and fibroblasts found in HIOs *in vitro* and following *in vivo* transplantation (Finkbeiner et al., 2015b; Spence et al., 2010; Watson et al., 2014). In this current work, we leveraged the plasticity of mesodermal progenitor cells early in HIO differentiation and demonstrated that a subset of these cells can be induced to differentiate into ECs and further expanded and maintained under the optimized vHIO protocol.

Through a scRNA-seq time course analysis of HIO development, we demonstrate that differentiation of HIO mesenchyme into vasculature takes place normally during early HIO differentiation, but that these cells are rare, and are mostly lost over time. Previous work has benchmarked d0 HIOs (intestinal spheroids) to embryonic day 8.5 (E8.5) mouse intestine (Spence et al., 2010). Vascularization of the gut is thought to begin a day later, around E9.5 (Hatch and Mukoyama, 2014), which follows a similar developmental progression as young HIOs, with ECs emerging within 72 hours of 3D culture. By targeting young HIOs with exogenous vascular induction and survival cues, we were able to increase the proportion and longevity of ECs within HIOs. While this is the first report demonstrating an

endogenous EC population within HIOs, EC populations have been previously described in human kidney organoids (van den Berg et al., 2018; Combes et al., 2019; Czerniecki et al., 2018; Freedman et al., 2015; Low et al., 2019; Takasato et al., 2015). Vasculature is a mesoderm derivative (Ferguson et al., 2005; Risau and Flamme, 1995), although the exact origins of intestinal and renal ECs are not fully understood. The robust mesoderm patterning that occurs in both organoid systems likely produces precursors for the native EC populations observed; however, recent pseudotime analysis performed on human kidney organoids suggests that ECs might be derived from a subset of mesodermal nephron progenitor cells that co-express *KDR* (Low et al., 2019). Similar to what we have observed in HIOs, ECs within human kidney organoids can be expanded with targeted growth factor modulation (Czerniecki et al., 2018; Low et al., 2019) or exposure to flow using microfluidics (Homan et al., 2019). Incorporation of flow into the vHIOs is an exciting future direction for this platform, as it might expand and enhance the stability the vascular networks and increase the appeal of this system for drug discovery.

Several studies have demonstrated that vasculature patterning is organ-specific (Daniel et al., 2018; Ding et al., 2010; Feng et al., 2019; Kalucka et al., 2020; Lee et al., 2014; Marcu et al., 2018; Nolan et al., 2013); however, whether or not organoids are programmed with this patterning information *in vitro* was not known. Furthermore, profiling of organ-specific ECs during development has not included human intestinal ECs prior to this work. Our data define a human intestinal EC signature that includes almost 150 genes that are enriched in intestinal ECs relative to lung and kidney ECs. Recent work described an adult murine EC atlas spanning 11 organ systems at single cell resolution, and provides a foundational resource to compliment the work presented here, aimed at understanding human intestinal EC patterning (Kalucka et al., 2020). Several of the validated human organ-specific markers identified in our dataset demonstrated similar expression patterns in adult mouse atlas, suggesting that organ-specific signatures established during development are retained into adulthood, although future work should confirm these findings in both developing and adult human and murine tissue. Further, a formal interrogation into mouse-human EC differences will yield an important understanding of species-specific differences. A unique strength of the current approach is the simultaneous characterization of organ-matched EC and non-EC populations. This strategy facilitated the identification of organ-specific EC-enriched signatures while at the same time understanding where and when these markers may be expressed in non-EC cell types in other organ systems.

While the vHIO platform described here constitutes the first *in vitro* organoid model known to contain appropriately patterned endogenous vasculature, it remains unknown what cell types and signals are responsible for inducing organ-specific gene expression and patterning in ECs. However, the scRNA-seq characterization of HIOs across developmental time provides an unprecedented insight into the diverse cell types present. Future work can leverage both scRNA-seq and spatial transcriptomics to interrogate this patterning and can leverage the modular nature of the HIO system to systematically test which cell type(s) and the molecular mechanisms that are responsible for inducing intestine-specific patterning in ECs. Better understanding these mechanisms will likely shed significant light on organ development. Reciprocal signaling has been shown to occur between ECs and the surrounding organ microenvironment during development (Kao et al., 2015; Lammert, 2001;

Lammert et al., 2003; Lazarus et al., 2011; Matsumoto et al., 2001; Vila Ellis et al., 2020) and ECs can influence development through the supply of membrane-bound or secreted factors, termed “angiocrine factors” (Rafii et al., 2016). Identifying these angiocrine roles of ECs has been challenging to study using *in vivo* animal models, as the vasculature is highly sensitive to modulations *in vivo* (Ferrara et al., 1996; Shalaby et al., 1995), and it is technically challenging to parse out unique angiocrine roles from metabolic requirements for the vasculature.

The vHIO model represents an *in vitro* system that can be used to study the dynamic EC-organ crosstalk during intestinal development. To this end, we performed *in silico* ligand-receptor analysis using CellphoneDB (Efremova et al., 2020) to speculate how ECs might be reciprocally communicating with HIO epithelial and mesenchymal lineages in addition to autocrine signaling. This analysis identified HIO ECs as a putative source of PDGFB, while the HIO mesenchyme expresses its receptor *PDGFRB*. PDGFB/PDGFRB signaling is crucial for pericyte recruitment during angiogenesis *in vivo* (Armulik et al., 2005, 2011; Gaengel et al., 2009). One possibility is that enhanced differentiation of ECs resulting from the vHIO protocol simultaneously improves pericyte differentiation and migration within the HIO mesenchyme through an increased supply of PDGFB. Future experiments can be designed to examine pericyte localization within vHIO in relation to the ECs. Additionally, ligand-receptor analysis identified HIO mesenchymal expression of ligand *CXCL12* (also known as SDF-1) and HIO EC expression of its receptor *CXCR4*. The CXCL12/CXCR4 signaling has been previously shown to be a key for proper intestinal vascularization (Ara et al., 2005; Tachibana et al., 1998). Taken together, these examples highlight that the HIO-EC interaction competency is consistent with physiologically relevant interactions observed in *in vivo* systems. Importantly, expression patterns identified here by *in silico* analysis should be validated *in situ*. Future work will continue exploring HIO-EC crosstalk, focusing on both how ECs are instructed to adopt organ-specific patterning, and also how ECs, in a perfusion independent context, might contribute to HIO development and maturation *in vitro*.

Taken together, our data identified an unexpected, rare population of EC-like cells that arise early during HIO differentiation. By developing a method that incorporates EC-inductive and maintenance growth factors, this population can be maintained for months *in vitro*. Through extensive transcriptional characterization and validation of organ-specific primary ECs across primary intestine, lung, and kidney during human development, we demonstrate the ECs co-differentiated within HIOs do undergo organ-specific patterning *in vitro*. vHIO improves both complexity and biological resemblance to the native developing intestine and comprises an amenable model to study EC-organ crosstalk during intestinal organogenesis.

METHODS

RESOURCE AVAILABILITY

Lead Contact—Contact Jason R. Spence at spencejr@umich.edu for requests for materials.

Materials Availability—This study did not generate new unique reagents.

Data Code and Availability—Sequencing data used in this study is deposited at EMBL-EBI ArrayExpress. Single cell RNA sequencing of human fetal lung, intestine and kidney: Human fetal lung (E-MTAB-8221) (Miller et al., 2020), Human fetal intestine (Czerwinski et al., 2020), and human fetal kidney – this study (E-MTAB-9083); HIO and vHIO – this study (E-MTAB-9228). Bulk RNA sequencing from FACS purified human fetal lung, intestine and kidney ECs and from cultured HUVECs – this study (E-MTAB-9372). Accession numbers for deposited data are also provided in the Key Resources Table. Code used to process raw data can be found at: <https://github.com/iason-spence-lab/Holloway2020>

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Human pluripotent stem cells—The University of Michigan Human Pluripotent Stem Cell Research Oversight Committee approved all experiments using human embryonic (ESC) and induced pluripotent stem cells (iPSC). vHIO and matched control HIOs were generated from 3 independent lines for these studies. vHIOs and control HIOs were generated from hESC line H9 (NIH registry no 0062) and iPSC lines WTC11 (Kreitzer et al., 2013) and 72.3 (McCracken et al., 2014).

Derivation of human intestinal organoids from hPSCs

Endoderm induction and intestinal spheroid generation: Differentiation of hPSCs into human intestinal spheroids (d0 HIOs) was carried out as previously described (Capeling et al., 2020; McCracken et al., 2011; Tsai et al., 2017). Briefly, hPSCs were patterned into definitive endoderm (DE) by supplementing Roswell Park Memorial Institute 1640 (RPMI-1640) with Activin A (100 ng/ml) for 3 days while increasing HyClone FBS concentration (0%, 0.2%, 2%) each day of DE induction. Following 3 days of DE induction, hindgut patterning was carried out through addition of FGF4 (500 ng/ml) and CHIR99021 (2 μ m) to RPMI-1640 containing 2% HyClone FBS. All medias used in differentiation contain 1x PenStrep. Media was changed daily, and spheroids were collected after 5 days of hindgut patterning. Spheroids were embedded in Matrigel (8 mg/ml, Corning, 354234). Spheroids were equally allocated between control and vHIO conditions for comparison studies.

Differentiation of control HIOs: Spheroids (d0 HIOs) were incubated in ENR media comprised of Minigut basal media, supplemented with EGF (100 ng/ml), Noggin (100 ng/ml), and R-Spondin-2 (5% conditioned media) for 3 days to pattern duodenal identity. After 3 days, ENR media was replaced with Minigut media containing only EGF (100 ng/ml). HIOs were analyzed between 14–28 days of culture. Minigut media is composed of the following components: Advanced DMEM:12 (Life Technologies, 12634), 1x B27 supplement (Life Technologies, 17504044), 1x GlutaMAX (Life Technologies, 35050061), 15 mM HEPES (Life Technologies, 15630080). All medias used in differentiation contain 1x PenStrep (Life Technologies, 15140). HIO maintenance and splitting was carried out as previously described with media changes every 3–4 days (Capeling et al., 2020; McCracken et al., 2011).

Differentiation of vHIOs: Spheroids (d0 HIOs) were incubated in ENR media comprised of Minigut basal media, supplemented with EGF (100 ng/ml), Noggin (100 ng/ml), and R-Spondin-2 (5% conditioned media) for 2 days to pattern duodenal identity. VEGF (50 ng/ml)

was added to the ENR media for the final 24 hours of duodenal patterning. The following day, ENR was replaced by Minigut containing EGF (100 ng/ml), VEGF (50 ng/ml), bFGF (25 ng/ml), and BMP4 (25 ng/ml) for 3 days. After these 3 days, vHIOs were grown in Minigut media supplemented with EGF (100 ng/ml) and VEGF (25 ng/ml) for the duration of culture. vHIO maintenance and splitting was carried out identically as control HIOs, as previously described with media changes every 3–4 days (Capeling et al., 2020; McCracken et al., 2011).

Primary tissue collection—Use of human tissue was reviewed and approved by The University of Michigan Institutional Review Board (IRB). De-identified human fetal lung, intestine, and kidney tissue was obtained from the University of Washington Laboratory of Developmental Biology. Tissue was shipped overnight in Belzer-UW Cold Storage Solution (ThermoFisher, NC0952695) with cold packs, as previously published (Menon et al., 2018; Miller et al., 2020).

METHOD DETAILS

Tissue fixation, paraffin processing and storage—Sample-matched human fetal lung, intestine, and kidney tissue samples were collected and processed into ~1–2 cm fragments. Tissues were fixed in 10% neutral buffered formalin (NBF) overnight (organoids) or 24 hours (primary tissue) at room temperature. The following day, tissues were washed with UltraPure Distilled Water (Invitrogen, 10977–015) for 3 changes for a total of 2 hours. Tissue was dehydrated in a methanol series (25%, 50%, 75%, 100%) diluted in UltraPure Distilled Water. Tissue was incubated for 60 minutes in each dehydration solution at room temperature. Tissue was stored long-term in 100% Methanol at 4°C. Prior to paraffin processing, tissue was equilibrated in 100% ethanol for an hour followed by 70% ethanol. Tissue was paraffin perfused using an automated tissue processor (Leica ASP300) with 1 hour solution changes overnight. Paraffin processed tissue was embedded and stored at room temperature with silica desiccant packets in a sealed container.

Immunofluorescence staining—Protein staining was conducted as previously described (Spence et al., 2009). Paraffin processed tissue was embedded and 5–7µm sections were cut. Paraffin sections were first deparaffinized in Histoclear and re-hydrated. Antigen retrieval was performed by steaming slides in a sodium citrate buffer for 20 minutes. Slides underwent a blocking step using the appropriate serum (5% serum in PBS + 0.1% Tween 20) for 1 hour at room temperature. Primary antibodies were diluted in blocking solution and slides were incubated with antibodies overnight at 4°C. The following day, slides were washed, and incubated with appropriate secondary antibodies diluted in a blocking buffer for 1 hour at room temperature together with DAPI (1µg/mL). Slides were washed and mounted using Prolong Gold (Thermo Fisher, P10144). A list of antibodies and dilutions can be found in the Key Resources Table and Supplementary Table 4.

Multiplex Fluorescent *In Situ* Hybridization (FISH)—Paraffin blocks were sectioned to generate 5 µm-thick sections. Sections were used within one week for optimal results, and the assay was carried out in RNase-free conditions by treating all materials with RNase-away (Molecular Bioproducts Inc., 7005–11) prior to use. Slides were stored at room

temperature in a sealed slide box with silica desiccant packets. Slides were baked for 1 hour in a 60°C dry oven a day prior to starting the procedure. The multiplex fluorescent *in situ* hybridization (FISH) protocol was performed according to the manufacturer's instructions (ACD; RNAscope Multiplex Fluorescent v2 manual protocol, 323100-USM) under standard antigen retrieval conditions and optimized protease treatment conditions for each tissue (lung 4 min., intestine 30 min., kidney 20min, HIO 20 min). Immunofluorescent protein staining was performed as previously described (Spence et al., 2009) on 5–7 µm sections. A list of RNAscope probes, TSA reagents, and antibodies can be found in the Key Resources Table and Supplementary Table 4. Each FISH stain was performed on at least 2 biological replicates. All imaging was done using a Leica SP5 or Nikon A1 confocal and images were assembled using Photoshop CC. Images were adjusted to optimize for visualization. For all images, any post-image processing (i.e. pseudocoloring, brightness, contrast, LUTs) was performed equally on entire images from a single experiment.

Wholemount staining of HIOs—HIOs were fixed for 30–60 minutes depending on size in 10% NBF at room temperature on a rocker. Samples were washed three times for 30-minutes in blocking solution (5% normal donkey serum in PBS with 0.1% triton). HIOs were transferred to permeabilization solution (PBS with 0.25% triton- spheroids or PBS with 0.5% triton- HIOs) for 30–60 minutes at room temperature, followed by three 30-minute washes in blocking solution and an additional 1 hour incubation in blocking solution. Primary antibodies against E-cadherin and CD144 were added to blocking solution and incubated for 24 hours at 4 °C (see key resources table for details). The following day, primary antibodies were removed and HIOs were subjected to three 30-minute washes in blocking solution. Appropriate fluorophore conjugated secondary antibodies were incubated with samples for 24 hours at 4 °C. The next day, secondary antibodies were removed and HIOs were washed three times for 30-minutes in blocking solution. DAPI (0.1 mg/ml) was added to the first. Samples were mounted onto slides containing secure-seal spacers (Invitrogen, S24737). Optical clearing was achieved by incubating HIOs in Focus Clear for 10–20 minutes at room temperature. This process was repeated with fresh Focus Clear until tissue was cleared. Focus Clear was replaced by Prolong Gold and slides were coverslipped. All imaging was done using a Nikon A1 confocal and images were assembled using Photoshop CC. Z-stack series (~0.8–1.25 µm steps) were captured and 3D rendering was performed using Imaris.

Isolation of endothelial cells from primary tissue—Primary human fetal tissue was dissociated into single cell suspensions for FACS isolation of ECs according to a previously published protocol (van Beijnum et al., 2008). Single cell suspensions were passed through a 70µm filter, pelleted, and resuspended in staining buffer comprised of PBS with 1x PenStrep, 2 µM EDTA, and 2% FBS. Cells were stained with CD144-APC and CD31-PE or corresponding isotype controls for 30 minutes on ice in a total volume of 100 µl per 1×10⁷ cells. Cells were washed three times in excess staining buffer accompanied by centrifugation at 300xg for 5 mins between washes. Cells were resuspended in staining buffer with DAPI (0.2 µg/ml). Antibody stained samples and controls were analyzed and sorted on a FACS Aria III cell sorter (BD), and analysis was performed using BD FACSDiva software.

Any post-acquisition analysis was performed using FlowJo. Cells were sorted into staining buffer, and after sorting cells were snap frozen and stored at -80°C prior to RNA isolation.

Flow cytometric analysis of HIOs—HIOs were removed from Matrigel droplets and transferred to an enzymatic solution. The enzyme solution is comprised of 1 ml dispase (2.5 units/mg) and 9 ml collagenase type II (0.1%) in PBS per 1 gram of tissue. Tissue digestions were carried out (~1 hour) at 37°C , agitating the solution every 10 minutes via stereological pipetting. Enzymatic reactions were quenched by adding 2x the volume of serum-containing (20% FBS) DMEM:F12 media. Cell suspensions were then passed through a $70\text{ }\mu\text{m}$ filter to remove any undigested clumps. Cells were spun down (400 g for 5 mins at 4°C) and resuspended in a staining buffer comprised of PBS with 1x PenStrep, and 2% FBS. Cells were counted using a hemacytometer, spun down (300 g for 10 minutes), and resuspended in appropriate volumes (100–200 μl) for antibody staining. Cell suspensions were stained with CD31-APC, CD144-PE, or corresponding isotype controls according to the manufacturers recommended dilution (see key resources table). Staining took place at 4°C for 10 minutes. Cell suspensions were washed by adding 2 mls of buffer, followed by centrifugation (300 g for 10 minutes). Pellets were resuspended in 500 μl of buffer, and DAPI (0.2 $\mu\text{g/ml}$) was added to appropriate staining conditions. Flow cytometric analysis was performed using a Sony MA900 cell sorter and accompanying software.

RNA isolation and Bulk RNAseq of primary ECs—RNA was isolated from snap frozen cell pellets using the RNeasy Mirco Kit (74004, Qiagen), according to manufacturer's guidelines. cDNA libraries were prepared using the SMARTer Stranded Total RNA-Seq Kit v2- Pico Input (634412, Takara). A total of 32 samples were sequenced for 50-bp single-end reads across 4 lanes on an Illumina HiSeq 2500 by the University of Michigan Advanced Genomics Core. Bulk RNA sequencing analysis was performed as previously described (Tsai et al., 2018). All reads were aligned to an index of transcripts from human genes within the Ensembl GRCh38 and quantified using Kallisto (Bray et al., 2016). Gene level data generated from Kallisto was used for TMM normalization in edgeR to create normalized data matrix of pseudocounts (Robinson et al., 2009). Principal component analysis and sample clustering were done in R using the 'cluster' and Bioconductor 'qvalue' packages (Storey et al., 2019). Genes were clustered by k-means clustering, using the KMeans function of the scikit learn package, after z-score transformation of pseudocount data (Pedregosa et al., 2011).

Single Cell Preparation of tissue for single cell RNA sequencing

Human Fetal Tissue: Cell dissociations were carried out similar to previously published methods (Miller et al., 2020). To dissociate human fetal tissue to single cells, tissue was mechanically minced into small fragments, and in a petri dish filled with ice-cold 1X HBSS (with Mg^{2+} , Ca^{2+}). This tissue was then transferred to a 15 mL conical tube. Dissociation enzymes and reagents from the Neural Tissue Dissociation Kit were used, and all incubation steps were carried out in a refrigerated centrifuge pre-chilled to 10°C unless otherwise stated. All tubes and pipette tips used to handle cell suspensions were pre-washed with 1% BSA in HBSS to prevent adhesion of cells to the plastic. Tissue was treated for 15 minutes at 10°C with Mix 1. Mix 2 was added to the digestion, and tissue was incubated for 10 minute

increments at 10°C until digestion was complete. After each 10 minute incubation, tissue was agitated using a P1000, and tissue digestion was visually assessed under a stereo microscope. This process continued until the tissue was fully digested. Cells were filtered through a 70 µm filter coated with 1% BSA in 1X HBSS, spun down at 500g for 5 minutes at 10°C and resuspended in 500µl 1X HBSS (with Mg^{2+} , Ca^{2+}). 1 mL Red Blood Cell Lysis buffer (Roche cat. No 11814389001) was then added to the tube and the cell mixture was placed on a rocker for 15 minutes at 4°C. Cells were spun down (500g for 5 minutes at 10°C), and washed twice by suspension in 2 mLs of HBSS + 1% BSA followed by centrifugation. Cells were counted using a hemocytometer, then spun down and resuspended (if necessary) to reach a concentration of 1000 cells/µL and kept on ice. Single cell droplets were immediately prepared on the 10x Chromium according to manufacturer instructions at the University of Michigan The Advanced Genomics Core, with a target of capturing 5,000–10,000 cells. Single cell libraries were prepared using the Chromium Next GEM Single Cell 3' Library Construction Kit v2 (lung and intestine samples) or v2 (kidney samples) according to manufacturer instructions.

Human Intestinal Organoids (d3, d7, and d14): To dissociate human intestinal organoids to single cells, organoids were mechanically isolated from Matrigel droplets and then tissue minced into small fragments using a scalpel in a petri dish filled with ice-cold 1X HBSS (with Mg^{2+} , Ca^{2+}). This tissue was then transferred to a 15 mL conical tube. Dissociation enzymes and reagents from the Neural Tissue Dissociation Kit (Miltenyi, cat. no. 130–092-628) were used, and all incubation steps were carried out in a 37°C incubator unless otherwise stated. All tubes and pipette tips used to handle cell suspensions were pre-washed with 1% BSA in HBSS to prevent adhesion of cells to the plastic. Tissue was treated for 15 minutes at 37°C with Mix 1. Mix 2 was added to the digestion, and tissue was incubated for 10 minute increments at 37°C until digestion was complete. After each 10 minute incubation, tissue was agitated using a P1000, and tissue digestion was visually assessed under a stereo microscope. This process continued until the tissue was fully digested. Cells were filtered through a 70 µm filter coated with 1% BSA in 1X HBSS, spun down at 500g for 5 minutes at 4°C and resuspended in 500µl 1X HBSS (with Mg^{2+} , Ca^{2+}). Cells were counted using a hemocytometer, then spun down and resuspended (if necessary) to reach a concentration of 1000 cells/µL and kept on ice. Single cell droplets were immediately prepared on the 10x Chromium according to manufacturer instructions at the University of Michigan The Advanced Genomics Core, with a target of capturing 5,000–10,000 cells. Single cell libraries were prepared using the Chromium Next GEM Single Cell 3' Library Construction Kit v2 according to manufacturer instructions.

Human intestinal spheroids (d0 HIOs): To dissociate human intestinal spheroids to single cells, spheroids were collected and transferred into a BSA coated (1 % BSA in 1x HBSS) 5ml conical. Excess media was removed, and digestion was carried out by incubating spheroids in TrypLE (2 ml) in a 37°C incubator for 10 minutes. Tissue was agitated using a P1000, digestion was visually assessed under a stereo microscope, and returned to the 37°C incubator for another 10 minutes. This process continued until the tissue was fully digested. Cells were filtered through a 70 µm filter coated with 1% BSA in 1X HBSS, spun down at 500g for 5 minutes at 4°C and resuspended in 500µl 1X HBSS (with Mg^{2+} , Ca^{2+}). Cells

were counted using a hemocytometer, then spun down and resuspended (if necessary) to reach a concentration of 1000 cells/ μ L and kept on ice. Single cell droplets were immediately prepared on the 10x Chromium according to manufacturer instructions at the University of Michigan The Advanced Genomics Core, with a target of capturing 5,000–10,000 cells. Single cell libraries were prepared using the Chromium Next GEM Single Cell 3' Library Construction Kit v2 according to manufacturer instructions.

RNA Extraction and qRT-PCR Analysis of HIOs—At least 3 wells containing 10–20 HIOs per well, for each biological replicate was collected for RNA extraction and qRT-PCR analysis. mRNA for qRT-PCR was isolated using the MagMAX-96 Total RNA Isolation Kit. RNA quality and concentration was determined on a Nanodrop 2000 spectrophotometer (Thermo Scientific). 100 ng of RNA for each sample was used to generate a cDNA library using Superscript VILO cDNA synthesis kit. qRT-PCR was performed on a Step One Plus Real-Time PCR System (Life Technologies) using SYBR Green Master Mix (Qiagen). Expression was calculated as a change relative to GAPDH expression using arbitrary units, calculated using the following equation: $[2^{-(\text{GAPDH Ct} - \text{Gene Ct})}] \times 1000$. Data were plotted as fold change of arbitrary expression of a treatment condition over a control condition. For this analysis, expression values for each gene for each sample, including controls, were divided by the average expression of that gene for the control group. Fold change was calculated as follows: $[\text{Expression of Gene in Treatment Condition} / \text{Average Expression of Gene in Controls}]$. A list of qRT-PCR primers used can be found in the Key Resources Table

QUANTIFICATION AND STATISTICAL ANALYSIS

All graphs and statistical tests were performed in GraphPad Prism 8 software. For analysis carried out in Figure 6, pooled control and vHIO organoids from one differentiation experiment were used for scRNA-seq, yielding $n=199$ individual HIO ECs for further analysis. A similarity score for fetal intestine, lung and kidney EC gene expression profiles was generated for each cell ($n=199$ cells per organ). To examine significance differences between groups, a one-way Analysis of Variance (ANOVA) was performed followed by Tukey's multiple comparisons analysis comparing the mean of each group to the mean of every other group. A p-value of less than 0.05 was considered significant. On graphs, p-values for multiple comparisons after ANOVAs are reported as followed: **** $p < 0.0001$.

scRNA-seq Data preprocessing, Cluster Identification and Cell Type Scoring—

All single-cell RNA-sequencing was performed with an Illumina Novaseq 6000 at the University of Michigan Advanced Genomics Core. Raw data was processed using the 10x Genomics Cell Ranger v2 or v3 pipeline using human reference genome (hg19) to generate gene expression matrices. Analysis was performed using the Scanpy Single Cell Analysis for Python toolbox described in previously (Wolf et al., 2018). To ensure input data were of high quality, filtering parameters for gene count range, unique molecular identifier (UMI) counts, and mitochondrial transcript fraction were imposed on each data set. After organ-specific quality filtering parameters were applied, all primary human data sets were combined for the remainder of preprocessing. Gene expression levels were log normalized, highly variable genes were extracted, and effects of UMI count and mitochondrial transcript

fraction variations were regressed out by linear regression. Gene expression values were z-transformed before samples were again separated by organ for downstream analysis. A graph-based clustering approach was performed using the top 10–11 principal components. Further dimensional reduction was done using the UMAP algorithm (McInnes et al., 2018), and cluster identification was performed as previously described (Blondel et al., 2008). Endothelial cell clusters were identified based on expression of canonical markers (i.e. *CDH5*, *KDR*) and computationally extracted and re-clustered.

For cell type scoring of major cell classes (epithelium, mesenchyme, endothelial, neuronal, immune), gene sets for each class were curated as previously described (Czerwinski et al., 2020; Miller et al., 2020) (Supplementary Table 2), and log normalized and z-transformed raw counts were summed to generate cell type scores. For visualization of these data, cell type scores were mapped onto UMAP embeddings. For organ-specific EC gene signatures (lung EC signature (IECs); intestinal EC signature (iECs); kidney EC signature (kECs) gene sets from the bulk RNAseq k-means clustering analysis were used. A given gene set was filtered to include the most highly enriched genes using genes with a log normalized z-score ≥ 1.8 . This filtering resulted in gene lists containing 137 (lung), 89 (intestine), and 50 (kidney) genes, respectively (Supplementary Table 2). For comparison of primary fetal EC gene sets to HIO ECs, HIO ECs were queried for expression of genes from each set, and the mean, log normalized and z-transformed raw counts for each gene signature set was determined (i.e. the mean expression of the IECs, iECs, and kECs sets were determined in HIO ECs). Cell scores were mapped onto the HIO EC embeddings.

Ligand-receptor analysis in HIOs—*In silico* ligand-receptor analysis was performed using the CellPhoneDB v2 interactive website (Efremova et al., 2020, cellphonedb.org) on the 59d control HIOs and vHIO datasets. Log transformed and normalized counts along with cluster identities were used as input for the analysis using standard parameters (10% threshold, 10 statistical iterations). For analysis we used values from the “significant_means.txt” output file (Supplementary Table 3).

TABLE FOR AUTHOR TO COMPLETE

Please upload the completed table as a separate document. **Please do not add subheadings to the** Key Resources Table. If you wish to make an entry that does not fall into one of the subheadings below, please contact your handling editor. (**NOTE:** For authors publishing in Current Biology, please note that references within the KRT should be in numbered style, rather than Harvard.)

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Highlights

- scRNA-seq reveals endothelial cells (EC) in early hPSC-derived intestinal organoids
- VEGF, BMP4, and FGF2 used to expand and maintain endogenous ECs
- RNA-seq identifies organ-enriched human intestine, lung, and kidney EC profiles
- Intestinal organoid ECs most closely resemble native intestinal ECs

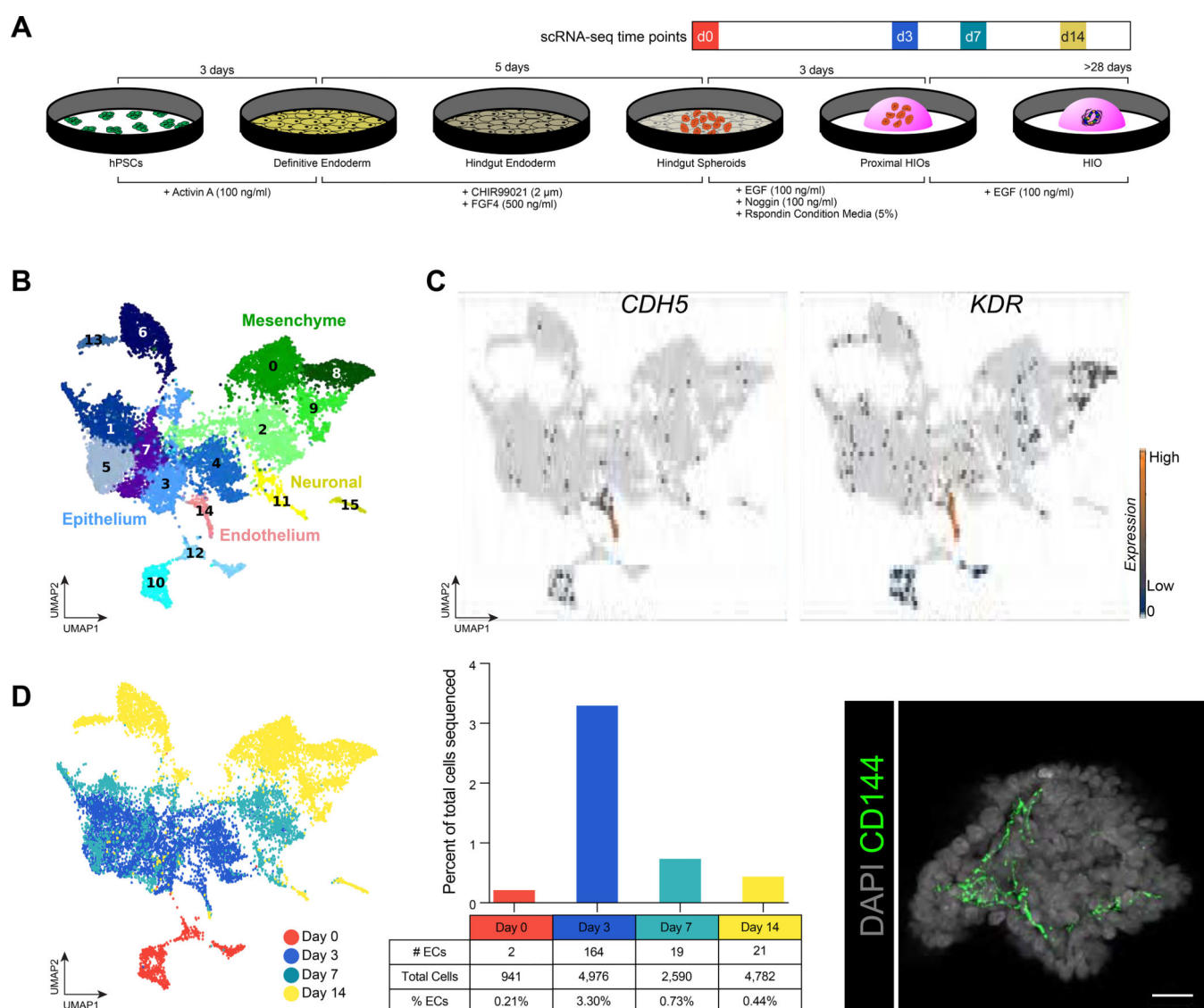


Figure 1. Identification of an endothelial cell-like population in HIOs.

(A) Overview of HIO differentiation protocol highlighting time points when HIOs were collected for single-cell RNA sequencing (scRNA-seq). (B) UMAP plot of 13,289 cells from all time points profiled with scRNA-seq predicted 16 cell clusters. Cluster identities were assigned based on expression of canonical lineage markers for epithelium, mesenchyme, endothelium, and neurons (see also Figure S1). (C) Feature plots for endothelial cell (EC) markers *CDH5* and *KDR* showing enrichment in Cluster 14. (D) UMAP plot illustrating the distribution of cells colored by sample (time point) and demonstrating the proportion of Cluster 14 (putative ECs) relative to total cells sequenced per timepoint (bar chart). (E) Wholemount immunofluorescent staining of d4 HIOs with the EC marker CD144 (green) and DAPI (grey). Representative data is shown for a single biological replicate. Scale bar represents 20 μ m.

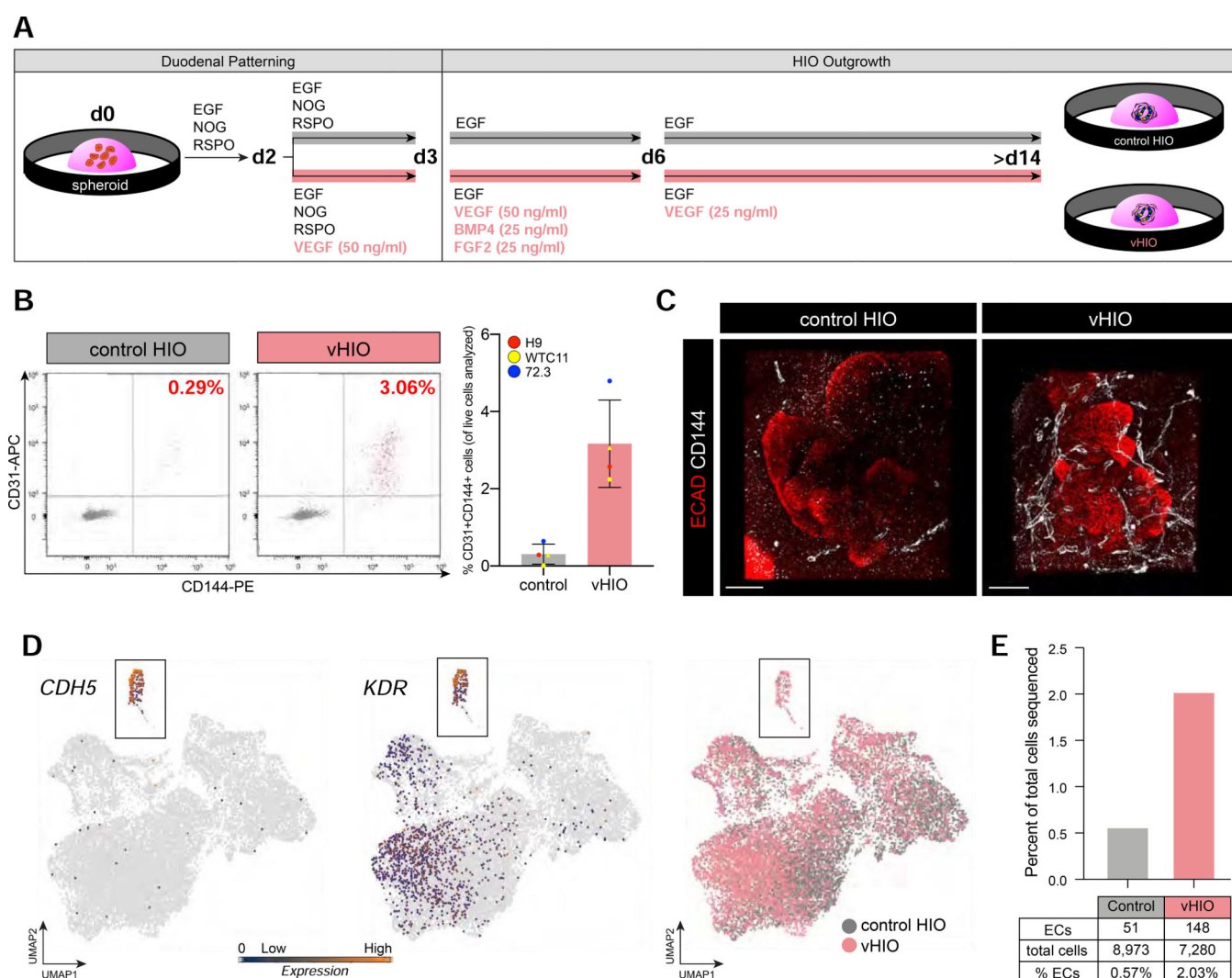


Figure 2. A method to increase survival of ECs in HIOs.

(A) Schematic of control and vHIO differentiation paradigms. **(B)** Representative flow cytometric plots of batch-matched 20d control HIOs and vHIOs for EC markers CD31 and CD144. Flow cytometric analysis of independent batch-matched differentiations of control HIO and vHIOs for presence of CD31+/CD144+ cells after 2–3 weeks in culture (n=4 differentiations in 3 different cell lines), showing an average ~9-fold increase of CD31+/CD144+ cells in the vHIOs. Error bars represent mean $\pm$ standard deviation (SD) (see also Figure S2). Color of data points corresponds to cell line used for differentiation. Matching shapes between control and vHIO conditions indicate batch-matched differentiations. **(C)** Maximum intensity projection of a wholemount confocal z-series staining for the EC marker CD144 (white) along with epithelial marker ECAD (red) in control HIOs and vHIOs (23d). Representative data is shown for a single biological replicate. Scale bar represents 100 μ m. **(D)** UMAP plots of 16,253 cells from scRNA-seq of HIOs (control and vHIO) showing expression of the EC markers *CDH5* and *KDR* and illustrating the distribution of cells colored by sample identity: grey – control HIO, pink – vHIO. Boxed region around the putative EC Cluster 5 (See also Figure S3). **(E)** Relative proportion of EC-like cells in

control HIOs and vHIO conditions compared to the total number of cells sequenced after 59d in culture.

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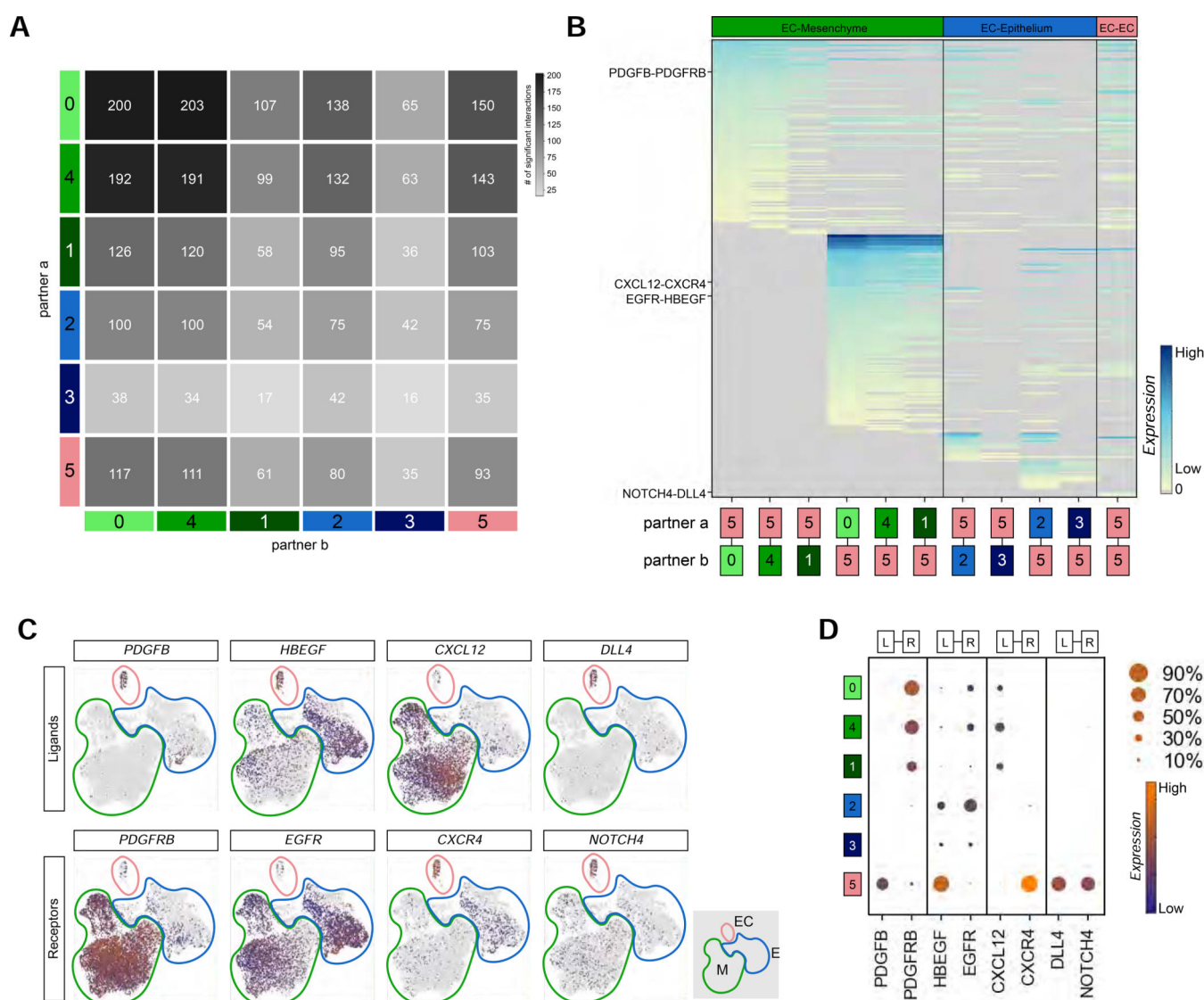


Figure 3. Ligand-receptor analysis focusing on ECs in control and vHIOs.

(A) Heatmap depicting the number of significant interactions ($p \leq 0.05$) identified between cell-cell pairings in 59d control and vHIOs using CellphoneDB. Numbers indicate the number of significant pairings between corresponding clusters. **(B)** Heatmap reflecting mean expression of significant ($p \leq 0.05$) interacting partners in indicated clusters. If a particular interaction was not statistically significant between two clusters, the mean expression was set to 0 (grey). **(C)** Feature plots showing expression of individual interacting partners for a subset of significant ligand-receptor interactions involving HIO ECs in 59d control HIOs and vHIOs, including: *PDGFB-PDGFRB*, *HBEGF-EGFR*, *CXCL12-CXCR4*, and *DLL4-NOTCH4*. Superimposed outlines demarcate epithelial (E-blue), mesenchymal (M-green), and endothelial (EC-pink) lineages. **(D)** Dotplot shows expression of selected ligand-receptor pairs across all clusters in 59d control HIOs and vHIOs. Dot size represents the proportion of cells in each cluster expressing a given marker, while color indicates log

normalized expression level. Colors are consistent with the cell type annotation presented in Figure S3– epithelium (2,3– blue), mesenchyme (0,1,4– green), and endothelium (5– pink).

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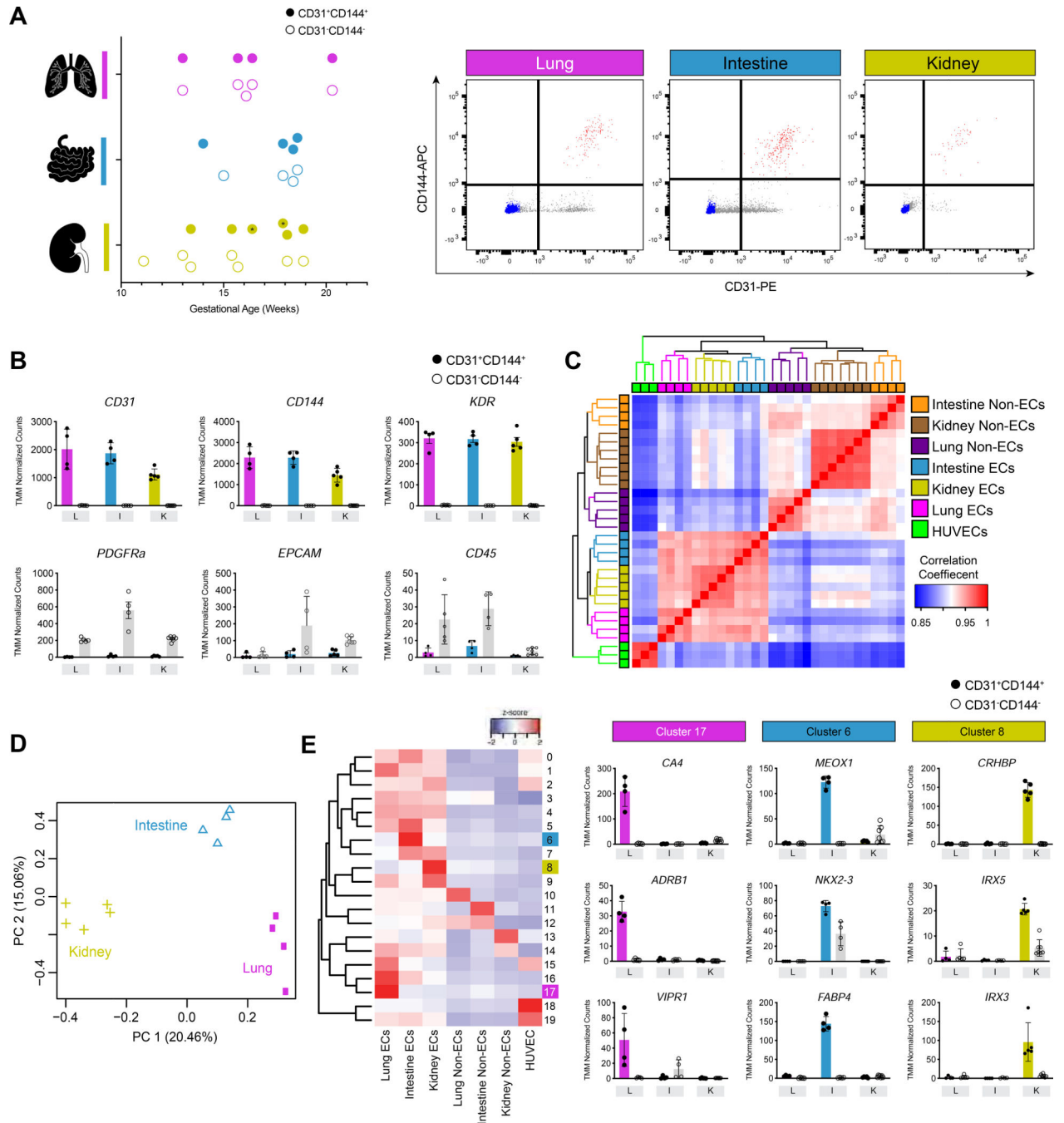


Figure 4. Defining organ-specific EC genes and gene signatures during human development.

(A) Left: Summary of all human lung, intestine, and kidney samples profiled by bulk RNAseq (n=4 lung EC, n=5 lung non-EC, n=4 intestine EC, n=4 intestine non-EC, n=5 kidney EC, and n=7 kidney non-EC biological replicates). Closed circles represent samples collected from sorted EC populations, and open circles represent sorted non-EC populations. The asterisks indicate two kidney EC samples isolated from biological specimens that were pooled prior to RNAseq. Samples are colored by organ system: Pink – lung; Blue – intestine; Yellow – kidney. Right: representative flow cytometry plots from each organ. ECs

(CD31+/CD144+, red) and non-ECs (CD31-/CD144-, blue) populations were isolated using FACS. **(B)** Bulk RNAseq data showing TMM normalized counts for EC genes (*CD31*, *CDH5*, *KDR*) highly enriched in the CD31+/CD144+ samples, and non-EC genes associated with the mesenchyme (*PDGFRa*), epithelial (*EPCAM*), and immune (*CD45*) transcripts enriched in CD31-/CD144- samples. Sample abbreviations are as followed: Lung (L), Intestine (I), and Kidney (K). Error bars represent mean $\pm$ standard deviation (SD). **(C)** Unsupervised hierarchical clustering of all primary human fetal samples profiled in this analysis, along with HUVECs. Each row corresponds to a biological replicate (except HUVEC samples, which represent n=3 technical replicates). **(D)** Principal component analysis of primary EC populations. **(E)** Left: K-means gene clustering (k=20, labeled 0–19) identifies organ-specific EC-enriched genes. Each row represents a cohort of genes that is enriched in one or more samples. Expression of the cohort is shown in each sample as the average normalized gene expression of all cohort genes in a given sample. Intestinal EC-enriched genes (Cluster 6 – blue box), Kidney EC-enriched genes (Cluster 8 – yellow box), Lung EC-enriched genes (Cluster 17 – pink box). Right: TMM normalized expression of representative organ-specific EC enriched gene candidates in human lung (pink: *CA4*, *ADRB1*, *VIRP1*), intestine (blue: *MEOX1*, *NKX2.3*, *FABP4*), and kidney (yellow: *CRHBP*, *IRX5*, *IRX3*) are shown across all primary samples. Error bars represent mean $\pm$ standard deviation (SD). Sample abbreviations are as followed: Lung (L), Intestine (I), and Kidney (K). Closed circles represent samples collected from sorted EC populations, and open circles represent sorted non-EC populations.

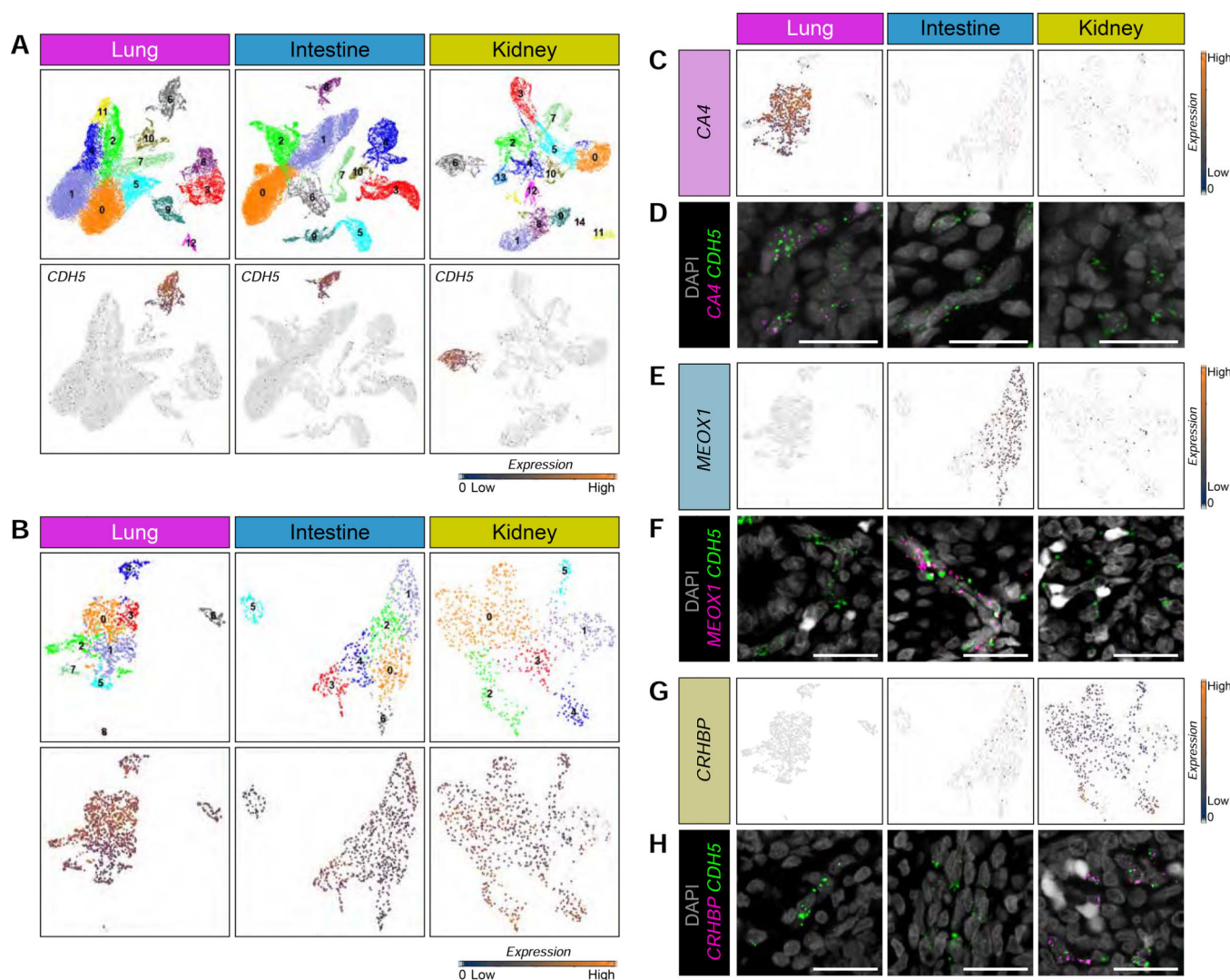


Figure 5. Validation of organ-specific EC-enriched genes.

(A) Top row: UMAP plots for human lung (Pink; n= 3; 26,501 cells), intestine (Blue; n=6; 26,010 cells), and kidney (Yellow; n=2; 9,535 cells) showing predicted cell clusters. Bottom row: Feature plots for the EC-specific marker *CDH5*, highlighting the EC population within each organ. (B) Top row: EC clusters from each organ were computationally extracted, re-clustered, and visualized using UMAP. A total of 1,361 lung ECs, 877 intestinal ECs, and 844 kidney ECs were included in the analysis. Bottom row: Feature plots of the EC marker *CDH5* among extracted clusters for each organ. (C-D) C: Feature plots of organ-specific EC genes (identified in Figure 4) showing expression in scRNA-seq data of the lung-specific EC candidate *CA4* in ECs of the lung, intestine and kidney. D: Multiplexed FISH for the pan-EC marker *CDH5* (green) and *CA4* (pink) in sample matched 16.7-week human fetal tissues. (E-F) E: Feature plot of scRNA-seq data of the intestine-specific EC candidate *MEOX1* expression among primary ECs. F: Multiplexed FISH for the pan-EC marker *CDH5* (green) and *MEOX1* (pink) in sample matched 16.7-week human fetal tissues. (G-H) G: Feature plots of scRNA-seq data of the kidney-specific EC candidate *CRHBP* expression among primary ECs. H: Multiplexed FISH for the pan-EC marker *CDH5* (green) and *CRHBP* (pink) in sample matched 16.7-week human fetal tissues.

(pink) in sample matched 17.1-week human fetal tissues. For all images, individual channels were consistently edited within an experiment for brightness/contrast by modifying the image look up table (LUT). Scalebars represent 25 μm .

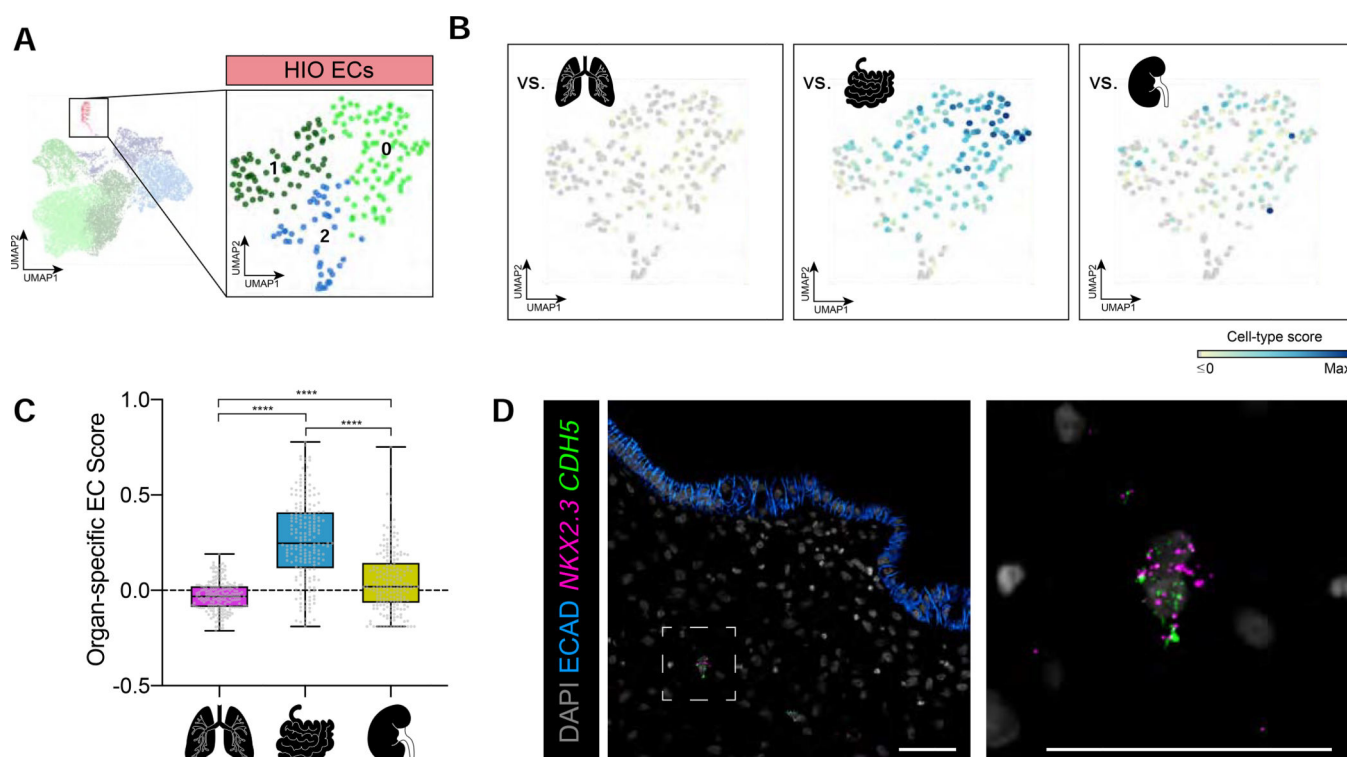


Figure 6. HIO ECs are transcriptionally similar to primary intestinal ECs.

(A) The EC cluster from Control HIO and vHIO scRNA-seq data (as shown in Figure 2) was computationally extracted, re-clustered and visualized using UMAP. **(B)** Expression of genes identified in gene signatures from primary human fetal ECs (as identified in Figure 3) were interrogated in ECs from HIOs (n=199 cells analyzed). An ‘organ-specific EC signature score’ was calculated as the average log normalized and z-transformed expression of organ-specific EC genes (from primary tissue gene signatures) in the ECs of the HIO (see Methods). The score, corresponding to the average expression of the signature gene sets in each cell, is plotted. Scores ≤ 0 were plotted as grey. **(C)** Box-and-whiskers plot of individual data points shown in (B) of the organ-specific endothelial cell type scoring. Statistical significance was calculated using a one-way ANOVA ($\alpha = 0.05$) followed by a Tukey post-hoc test. The intestine EC score was statistically different than both lung and kidney EC scores ($p < 0.001$). **(D)** FISH in an vHIO for the intestine-specific marker *NKX2.3* (pink) and the pan-EC marker *CDH5* (green), along with protein staining for ECAD (blue), and DAPI (grey). Representative data is shown for a single biological replicate. Individual channels were edited for brightness/contrast by modifying the image look up table (LUT). Scalebars represent 50 μm .

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Mouse anti E-Cadherin	BD Biosciences	Cat#610181; RRID:AB_397580
Goat anti-E-Cadherin	R&D	Cat#AF748; RRID:AB_355568
Rabbit anti-CD31	Atlas Antibodies	Cat#HPA004690; RRID:AB_1078462
Mouse anti-CD144	R&D	Cat#MAB9381; RRID:AB_2260374
Rabbit anti-FABP4	Sigma	Cat#HPA002188; RRID:AB_1078822
CD31-PE	Miltenyi	Cat#130-098-173; RRID:AB_2660561
CD144-APC	Miltenyi	Cat#130-100-709; RRID:AB_2655154
REA-APC isotype	Miltenyi	Cat#130-104-630; RRID:AB_2661703
mIgG1-PE	Miltenyi	Cat#130-098-845; RRID:AB_2661716
CD144-PE	Miltenyi	Cat#130-118-495; RRID:AB_2751528
REA Control (S)-APC	Miltenyi	Cat#130-113-434; RRID:AB_2733447
REA Control (S)-PE	Miltenyi	Cat#130-113-438; RRID:AB_2733893
CD31-APC	Miltenyi	Cat#130-117-314; RRID:AB_2727917
Donkey anti-mouse 488	Jackson ImmunoResearch	Cat#715-545-150; RRID:AB_2340846
Donkey anti-mouse Cy3	Jackson ImmunoResearch	Cat#715-165-150; RRID:AB_2340813
Donkey anti-mouse 647	Jackson ImmunoResearch	Cat#715-605-150; RRID:AB_2340862
Donkey anti-rabbit 488	Jackson ImmunoResearch	Cat#711-545-152; RRID:AB_2313584
Donkey anti-rabbit Cy3	Jackson ImmunoResearch	Cat#711-165-152; RRID:AB_2307443
Donkey anti-rabbit 647	Jackson ImmunoResearch	Cat#711-605-152 RRID:AB_2492288
Donkey anti-goat 488	Jackson ImmunoResearch	Cat#705-545-147; RRID:AB_2336933
Donkey anti-goat Cy3	Jackson ImmunoResearch	Cat#705-165-147; RRID:AB_2307351
Donkey anti-goat 647	Jackson ImmunoResearch	Cat#705-605-147; RRID:AB_2340437
Biological Samples		
Human Fetal Lung	University of Washington Laboratory of Developmental Biology	N/A
Human Fetal Intestine	University of Washington Laboratory of Developmental Biology	N/A
Human Fetal Kidney	University of Washington Laboratory of Developmental Biology	N/A
Chemicals, Peptides, and Recombinant Proteins		
EGF	R&D	Cat#236-EG
Noggin	R&D	Cat#6057-NG
VEGF	R&D	Cat#293-VE
Activin A	R&D	Cat#338-AC
bFGF	R&D	Cat#233-FB
BMP4	R&D	Cat#314-BP

REAGENT or RESOURCE	SOURCE	IDENTIFIER
CHIR99021	Tocris	Cat#4423
FGF4	Purified in house	(Sugawara et al., 2014)
B27 supplement	Life Technologies	Cat#17504044
HEPES	Life Technologies	Cat#15630080
GlutaMAX	Life Technologies	Cat#35050061
Dispase	Life Technologies	Cat#17105-041
Collagenase Type II	Life Technologies	Cat#17101-015
Red Blood Cell Lysis Buffer	Roche	Cat#11814389001
Critical Commercial Assays		
Neural Tissue Dissociation Kit (P)	Miltenyi	Cat#130-092-628
RNAscope Multiplex Fluorescent Reagent Kit v2	ACD	Cat#323100
SuperScript VILO cDNA Synthesis Kit	ThermoFisher	Cat#11754250
MagMAX-96 Total RNA Isolation Kit	Ambion	Cat#AM1830
QuantiTect SYBR Green PCR Kit	Qiagen	Cat#204145
SMARTer Stranded Total RNA-Seq Kit v2- Pico Input	Takara	Cat#634412
Chromium Next GEM Single Cell 3' Library Construction Kit v3	10x Genomics	Cat#PN-1000075
Chromium Next GEM Single Cell 3' Library Construction Kit v2	10x Genomics	Cat#PN-120237
Deposited Data		
Raw scRNAseq data (human fetal lung)	(Miller et al., 2020)	E-MTAB-8221
Raw scRNAseq data (human fetal kidney)	This paper	E-MTAB-9083
Raw scRNAseq data (HIO and vHIOs)	This paper	E-MTAB-9228
Raw scRNAseq data (human fetal intestine)	(Czerwinski et al., 2020)	<i>In progress</i>
Raw scRNAseq data (human fetal intestine and lung)	This paper	E-MTAB-9363
Raw bulk RNAseq data (human fetal ECs)	This paper	E-MTAB-9372
Experimental Models: Cell Lines		
HUVECs	ATCC	Cat#PCS-100-010 (lot 64278987)
H9	WiCell	Cat#WA09; RRID:CVCL_9773
WTC11	Cornell Institute	Cat#GM25256; RRID:CVCL_Y803
iPSC 72.3	CCHMC	(McCracken et al., 2014)
Oligonucleotides		
CTCTGCTCCTCCTGTTTCGAC (GAPDH-forward)	IDT	N/A
TTAAAAGCAGCCCTGGTGAC (GAPDH-reverse)	IDT	N/A
GCTTAGCCTCGTCGATGAAC (SOX2-forward)	IDT	N/A
AACCCCAAGATGCACAACCTC (SOX2-reverse)	IDT	N/A
CATCTTCCCAGGAGGAACAG (CD144-forward)	IDT	N/A
AGAGCTCCACTCACGCTCAG (CD144-reverse)	IDT	N/A
GCTGACCCTTCTGCTCTGTT (CD31-forward)	IDT	N/A
TGAGAGGTGGTGCTGACATC (CD31-reverse)	IDT	N/A

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms		
Prism 8.3.0	GraphPad	https://www.graphpad.com/scientific-software/prism/
Scanpy	(Wolf et al., 2018)	https://github.com/theislab/scanpy
Cellranger	10x Genomics	https://support.10xgenomics.com/single-cell-gene-expression/software/pipelines/latest/what-is-cell-ranger
Python 3.7.3	Python	Python.org
FlowJo	BD	https://www.flowjo.com/
CellphoneDB	(Efremova et al., 2020)	Cellphonedb.org
FACSDiva	BD	https://www.bdbiosciences.com/en-us/instruments/research-instruments/research-software/flow-cytometry-acquisition/facsdiva-software
Detailed methods and code for scRNAseq analysis	GitHub	https://github.com/jason-spence-lab/Holloway2020
Other		
Matrigel	Corning	Cat#354234
RNase AWAY	Molecular Bioproducts Inc.	Cat#7005–11
FocusClear	CelExplorer	Cat#FC-101
Histoclear II	National Diagnostics	Cat#HS-202
RNAscope Probe Hs-CDH5	ACD	Cat#437451-C2
RNAscope Probe Hs-ADRB1	ACD	Cat#469511
RNAscope Probe Hs-CA4	ACD	Cat#438561
RNAscope Probe Hs-CRHBP	ACD	Cat#573411
RNAscope Probe Hs-IRX3	ACD	Cat#588401
RNAscope Probe Hs-IRX5	ACD	Cat#441771
RNAscope Probe Hs-MEOX1	ACD	Cat#564321
RNAscope Probe Hs-NKX2–3	ACD	Cat#581651
RNAscope Probe Hs-VIPR1	ACD	Cat#566991
TSA Plus Cyanine 3 (1:1500–2000)	Akoya Biosciences	Cat#NEL744001KT
TSA Plus Cyanine 3 (1:1500–2000)	Akoya Biosciences	Cat#NEL745E001KT